



## CS504-Software Engineering -1

LATEST SOLVED SUBJECTIVES  
FROM MIDTERM PAPERS

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**Latest Subjective**

### CS504-MIDTERM SOLVED SUBJECTIVE WITH REFERENCES



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# CS504 Subjective

## Midterm Examination 2011

**Question No: 1 ( Marks: 3 )**

**Define Asynchronous Messages and Synchronous Messages.**

**Answer:- (Page 109)**

**Asynchronous Messages:-**

Asynchronous messages are “signals,” denoted by a half arrow. They do not block the caller. Asynchronous messages typically perform the following actions:

- ❖ Create a new thread
- ❖ Create a new object

Communicate with a thread that is already running

**Synchronous Messages:-**

Synchronous messages are “call events” and are denoted by the full arrow. They represent nested flow of control which is typically implemented as an operation call. In case of a synchronous message, the caller waits for the called routine to complete its operation before moving forward.

**Question No: 2 ( Marks: 5 )**

**Law of balancing act in software**

**Answer:- (Page 7)**

**The Balancing Act!**

Software Engineering is actually the balancing act. You have to balance many things like cost, user friendliness, Efficiency, Reliability etc. You have to analyze which one is the more important feature for your software is it reliability, efficiency, user friendliness or something else. There is always a trade-off among all these requirements of software. It may be the case that if you try to make it more user-friendly then the efficiency may suffer. And if you try to make it more cost-effective then reliability may suffer. Therefore there is always a trade-off between these characteristics of software. These requirements may be conflicting. For example, there may be tension among the following:

- ❖ Cost vs. Efficiency
- ❖ Cost vs. Reliability
- ❖ Efficiency vs. User-interface

A Software engineer is required to analyze these conflicting entities and tries to strike a balance.

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## Midterm Examination 2011

**Question No: 3 ( Marks: -- )**

**HOW DO YOU DETERMINE THAT AN OBJECTIVE BELONGS TO CERTAIN CLASS?**

**Answer:- (Page 85)**

The basic unit of object oriented design is an object. An object can be defined as a tangible entity that exhibits some well defined behavior. The structure and behavior of similar objects are defined in their common class. A class specifies an interface and defines an implementation.

**Question No: 4 ( Marks: -- )**

**What is meant by “System’s Static View”?**

**Answer:- (Software engineering ---Page 608 )**

**Static view of semantic classes.**

Requirements are assessed and classes are extracted (and represented) as part of the analysis model.

**Static view of attributes.** Every class must be explicitly described. The attributes associated with the class provide a description of the class.

**Static view of relationships.**

Objects are “connected” to one another in a variety of ways. The analysis model must represent these.

**Static view of behaviors.**

The relationships just noted define a set of behaviors that accommodate the usage scenario (use-cases) of the system

**Question No: 5 ( Marks: -- )**

**What is behavior driven perceptive of an objective?**

**Answer:- (Page 85)**

Behavior is how an object acts and reacts in terms of its state changes and message passing. The behavior of an object is completely defined by its actions. A message is some action that one object performs upon another in order to elicit a reaction. The operations that clients may perform upon an object are called methods.

**Question No: 6 ( Marks: -- )**

**What is Textual Analysis? Explain it**

**Answer:- (Page 90)**

Textual analysis was developed by Abbot and then extended by Graham and others. In this technique different parts of speech are identified within the text of the specification and these parts are modeled using different components.

**Question No: 7 ( Marks: -- )**

**What is Software Architecture?**

**Answer:- (Page 116)**

Architecture is the organizational structure of a system. Architecture can be recursively decomposed into parts that interact through interfaces, relationships that connect parts, and constraints for assembling parts. Parts that interact through interfaces include classes, components and subsystems.

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**Question No: 8 ( Marks: -- )**

**What is the Cardinality of Objective?**

**Answer :-** (Software engineering ---Page 334 )

**Cardinality.** The data model must be capable of representing the number of occurrences Objects in a given relationship. Tillmann [TIL93] defines the *cardinality* of an object/relationship pair in the following manner:

- ❖ One-to-one (1:1)—An occurrence of [object] 'A' can relate to one and only one occurrence of [object] 'B,' and an occurrence of 'B' can relate to only one occurrence of 'A.'
- ❖ One-to-many (1:N)—One occurrence of [object] 'A' can relate to one or many occurrences of [object] 'B,' but an occurrence of 'B' can relate to only one occurrence of 'A.' For example, a mother can have many children, but a child can have only one mother.
- ❖ Many-to-many (M:N)—An occurrence of [object] 'A' can relate to one or more occurrences of 'B,' while an occurrence of 'B' can relate to one or more occurrences of 'A.' For example, an uncle can have many nephews, while a nephew can have many uncles.

## Midterm Examination 2011

**Question No: 9 (Marks: 2)**

**Define abstraction?**

**Answer:-** (Page 79)

An abstraction is a technique in which we construct a model of an entity based upon its essential characteristics and ignore the inessential details.

**Question No: 10 (Marks: 2)**

**Is the design of software architecture a creative process?**

**Answer:-** (Page 120)

Design of software architecture is a creative and iterative process. This involves performing a number of activities, not necessarily in any particular order or sequence.

**Question No: 11 (Marks: 2)**

**Suppose you are working as a software engineer involved in the development of an e-commerce website. What are the 2 most important characteristics your software must have?**

**Answer:-** (Software engineering ---Page 797 )

Two most important characteristics are that software should be reliable and useable.

**Question No: 12 (Marks: 3)**

**What is the purpose of collaboration diagrams?**

**Answer:-** (Page 111)

Collaboration diagrams can also be used to depict the dynamic behavior of a system. They show how objects interact with respect to organizational units (boundaries!). Collaboration diagrams can also show synchronous, asynchronous, create, and destroy message using the same notation as used in sequence diagrams.

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**Question No: 13 ( Marks: 3 )****What is the difference between Aggregation and Association?****Answer:-** (Page 87)

As compared to association, aggregation implies a tighter coupling between the two objects which are involved in this relationship. Therefore, one way to differentiate between aggregation and association is that if the two objects are tightly coupled, that is, if they cannot exist independently, it is an aggregation, and if they are usually considered as independent, it is an association.

**Question No: 14 (Marks: 5)****what parameters are used to measure and analyze design quality?****Answer:-** (Page 71)

A software design can be looked at from different angles and different parameters can be used to measure and analyze its quality. These parameters include efficiency, compactness, reusability, and maintainability. A good design from one angle may not seem to be suitable when looked from a different perspective. For example, a design that yields efficient and compact code may not be very easy to maintain. In order to establish whether a particular design is good or not, we therefore have to look at the project and application requirements.

## Midterm Examination 2011

**Question No: 15 ( Marks: 5 )****Discuss some of the purpose of interacting diagram?****Answer:-** [Click here for detail](#)

The purposes of interaction diagrams are to visualize the interactive behaviour of the system. Now visualizing interaction is a difficult task. So the solution is to use different types of models to capture the different aspects of the interaction.

That is why sequence and collaboration diagrams are used to capture dynamic nature but from a different angle.

So the purposes of interaction diagram can be describes as:

- ❖ To capture dynamic behaviour of a system.
- ❖ To describe the message flow in the system.
- ❖ To describe structural organization of the objects.
- ❖ To describe interaction among objects.

**Question No: 16 (Marks: 5)****What should be consideration for maintain design?****Answer:-** (Page 71)

In order to make a design that is maintainable, it should be understandable and the changes should be local in effect. That is, it should be such that a change in some part of the system should not affect other parts of the system. This is achieved by applying the principles of modularity, abstraction, and separation of concern. If applied properly, these principles yield a design that is said to be more cohesive and loosely coupled and thus is easy to maintain.

**Question No: 17 ( Marks: 3 )****Purpose of collaborating diagram?****Answer:-** (repeated)

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**Question No: 18 ( Marks: 3 )**

**It is fact that good design makes maintenance easier. Which design principle help this to be achieved?**

**Answer:- (Page 71)**

A good design from one angle may not seem to be suitable when looked from a different perspective. For example, a design that yields efficient and compact code may not be very easy to maintain. In order to establish whether a particular design is good or not, we therefore have to look at the project and application requirements.

## **Midterm Examination 2011**

**Question No: 20 ( Marks: -- )**

**To manage the complexity of the system we need to apply the principle of abstraction. Discuss briefly?**

**Answer:- (Page 79)**

An abstraction is a technique in which we construct a model of an entity based upon its essential characteristics and ignore the inessential details. The principle of abstraction also helps us in handling the inherent complexity of a system by allowing us to look at its important external characteristic, at the same time, hiding its inner complexity. Hiding the internal details is called encapsulation.

**Question No: 21 (Marks: 2)**

**Differentiate between architectural design and system architecture in a single line?**

**Answer:- [Click here for detail](#)**

Architecture faces towards strategy, structure and purpose, towards the abstract while Design faces towards implementation and practice, towards the concrete.

**Question No: 5 ( Marks: 3 )**

**What are architectural designs Process, explain briefly?**

**Answer:- (Page 79)**

**System structuring: -**

System structuring is concerned with decomposing the system into interacting sub-systems. The system is decomposed into several principal sub-systems and communications between these sub-systems are identified.

**Control modeling:-**

Control modeling establishes a model of the control relationships between the different parts of the system.

**Modular decomposition:-**

During this activity, the identified sub-systems are decomposed into modules. This design process is further elaborated in the following section where architectural views are discussed.

**Question No: 5 ( Marks: 5 )**

**What should be the consideration for maintainable design?**

**Answer:- (rep)**

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**Question No: 5 ( Marks: 5 )**

**Discuss the relationship between sequence diagram and logical complexity?**

**Answer:- (Page 110)**

It is important to understand that the diagrams are meant to make things clear. Therefore, in order to keep them simple, special attentions should be paid to the conditional logic. If it is simple then there is no harm in adding it to the diagram. On the other hand if the logic is complex then we should draw separate diagrams like flow charts.

## **MIDTERM EXAMINATION - Spring 2010**

**Question No: 17 ( Marks: 2 )**

**To manage the complexity of the system we need to apply the principles of abstraction. Discuss briefly.**

**Answer: rep**

**Question No: 18 ( Marks: 2 ) p2chapter 5**

**Define Motivation of GUI**

**Answer:**

**Motivation for GUI**

- ❖ System users often judge a system by its interface rather than its functionality
- ❖ A poorly designed interface can cause a user to make catastrophic errors
- ❖ Poor user interface design is the reason why so many software systems are never used

**Question No: 19 ( Marks: 2 )**

**Is the design of software architecture a creative process?**

**Answer: rep**

**Question No: 20 ( Marks: 3 )**

**Keeping in mind the Connie's case study, what rule of thumbs was identified, list them down.**

**Answer:- (Page 101)**

**Who I Know - Rules of Thumb**

- ❖ an actor knows about its participants person knows about cashier
- ❖ a transaction knows about its participants a session knows about its register and cashier
- ❖ A transaction contains its transaction line items sale contains its sales line items
- ❖ A transaction knows its sub transactions session knows about its sales sale knows about its payments
- ❖ A place knows about its transactions store knows about its sessions
- ❖ A place knows about its descriptive objects store knows about its tax categories
- ❖ A container knows about its contents a store knows about its cashiers, items, and registers

**Question No: 21 ( Marks: 3 )**

**The CPU of a computer consists of an ALU and a CU. Intel and AMD are two types of CPUs, which are widely used in computers. Users of computers can be either novices or experts.**

**Consider the following classes.**

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- (i) CPU and AMD
- (ii) User and Computer
- (iii) CPU and ALU

**Answer:**

**CPU and AMD:-**

There is relationship between CPU and AMD is “A kind of” type as AMD is a kind of CPU.

**User and Computer:-**

The relationship between User and Computer is “simple Association” type as there is weak relationship between user and computer.

**CPU and ALU:-**

Relationship between CPU and ALU is “composition” type as ALU is a part of CPU.

**Question No: 22 ( Marks: 5 )**

**What is action-oriented approach for Software Design?**

**Answer: (Page 80)**

In the case of action-oriented approach, data is decomposed according to functionality requirements. That is, decomposition revolves around function. In the OO approach, decomposition of a problem revolves around data. Action-oriented paradigm focuses only on the functionality of a system and typically ignores the data until it is required. Object-oriented paradigm focuses both on the functionality and the data at the same time. The basic difference between these two is decentralized control mechanism versus centralized control mechanism respectively. Decentralization gives OO the ability to handle essential complexity better than action-oriented approach.

**Question No: 23 ( Marks: 5 )**

**Collaboration Diagrams depict Dynamic behavior of the system, explain it.**

**Collaboration diagrams:- Answer: (Page 110)**

Collaboration diagrams can also be used to depict the dynamic behavior of a system. They show how objects interact with respect to organizational units (boundaries!).

Since a boundary shapes communication between system and outside world e.g. user interface or other system, collaboration diagrams can be used to show this aspect of the system. The sequence of messages determined by numbering such as 1, 2, 3, 4, This shows which operation calls which other operation.

## **MIDTERM EXAMINATION**

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**Question No: 17 ( Marks: 2 )**

**To manage the complexity of the system we need to apply the principles of separation of concern. Discuss briefly**

**Answer: (Page 69)**

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Separation of concern allows us to deal with different individual aspects of a problem by considering these aspects in isolation and independent of each other. A complex system may be divided into smaller pieces of lesser complexity called modules.

**Question No: 18 ( Marks: 2 )**

**What is elaborated Use case? Explain it**

**Answer: (Page 36)**

After the derivation of the use case model, each use is elaborated by adding detail of interaction between the user and the software system. An elaborated use case has the following components:

Use Case Name, actors, summary, precondition, post-condition, extend, uses, normal course of events, alternative path, exception, assumption.

**Question No: 19 ( Marks: 2 )**

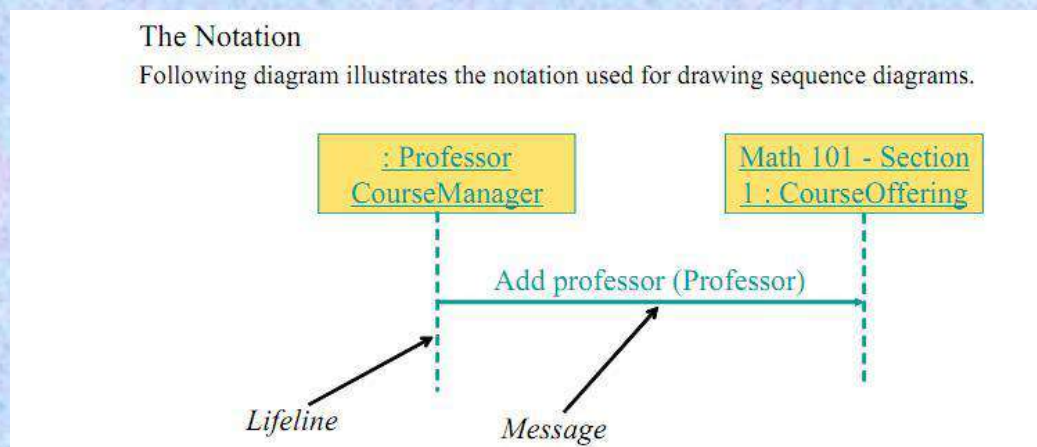
**What is Software Architecture?**

**Answer: rep**

**Question No: 20 ( Marks: 3 )**

**What notation is used for Sequence Diagrams? Draw it graphically.**

**Answer: (Page 106)**



**Question No: 21 ( Marks: 3 )**

**How can we decrease Coupling explain it.**

**Answer: (Page 113)**

That is, we can reduce the coupling of a system by minimizing the number of messages in the protocol of a class.

**Question No: 22 ( Marks: 5 )**

**Narrate some of the architectural design guidelines that can help in addressing non-functional requirements challenges.**

**Answer: ( Page 120 )**

Software architecture must address the non-functional as well as the functional requirements of the software system. Following are some of the architectural design.

**Performance** –Performance can be enhanced by localizing operations to minimize sub-system communication.

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That is, try to have self-contained modules as much as possible so that inter-module communication is minimized.

**Security** –Security can be improved by using a layered architecture with critical assets put in inner layers.

**Safety** –Safety-critical components should be isolated

**Availability** –Availability can be ensured by building redundancy in the system and having redundant components in the architecture.

**Maintainability**–Maintainability is directly related with simplicity. Therefore, maintainability can be increased by using fine-grain, self-contained components.

**Question No: 23 ( Marks: 5 )**

**What is the importance of Classification in identifying Classes and objects?**

**Answer: (Page 763)**

Consider a large university library. Tens of thousands of books, periodicals, and other information resources are available for use. But to access these resources, a categorization scheme must be developed. To navigate this large volume of information, librarians have defined a classification scheme that includes a Library of Congress classification code, keywords, author names, and other index entries. All enable the user to find the needed resource quickly and easily.

**Question No: 17 ( Marks: 2 )**

**Define abstraction**

**Answer: rep**

**Question No: 18 ( Marks: 2 )**

**What is Software Architecture?**

**Answer: rep**

**Question No: 19 ( Marks: 2 )**

**Suppose you are working as a software engineer involved in the development of an e-commerce website. What are the 2 most important characteristics your software must have?**

**Answer: rep**

**Question No: 20 ( Marks: 3 )**

**What is the purpose of collaboration diagrams?**

**Answer: rep**

**Question No: 21 ( Marks: 3 )**

**It is a fact that a good design makes maintenance easier. List a design principle which helps this to be achieved.**

**Answer: rep**

**Question No: 22 ( Marks: 5 )**

**Discuss the relationship between Sequence diagrams and logical complexity**

**Answer: rep**

**Question No: 23 ( Marks: 5 )**

**What is the difference between Association and composition?**

**Answer:- (OOP, PAGE 51-52)**

In Association, interacting objects have no intrinsic relationship with other object. It is the weakest link between objects. While in Composition An object may be composed of other smaller objects, the relationship between the “part” objects and the “whole”.

## **MIDTERM EXAMINATION**

**Spring 2010**

**CS504- Software Engineering – I**

**Question No: 23 ( Marks: 3 )**

**How can we decrease coupling? Explain it.**

**Answer: rep**

**Question No: 23 ( Marks: 5 )**

**What is meant by “system’s static View”? Discuss briefly.**

**Answer: rep**

**Question No: 23 ( Marks: 5 )**

**How the objects are identified in peter codd’s technique?**

**Answer: (Page 93)**

Objects are identifying in the following way.

### **Select actors**

Actors are people and organizations that take part in the system under consideration. Examples of actors are: person, organization (agency, company, corporation, foundation).

### **Select Participants**

A participant is a role that each actor plays in the system under consideration. Examples of participants are: agent, applicant, buyer, cashier, clerk, customer, dealer, and distributor. Etc.

### **Select Places**

Places are where things come to rest or places that contain other objects. Examples of places are: airport, assembly-line, bank, city, clinic, country, depot, garage and hospital etc.

### **Select Transactions**

Transactions are the “events”. These transactions usually come from a window (GUI), some object which monitors for significant event and logs that information, or a another system that interacts with the system under consideration and logs some information. Examples of transactions are: agreement, assignment, authorization, contract, delivery, deposit, incident, inquiry, order, payment, problem report, purchase and sales etc.

### **Select Container Objects**

Containers are objects that hold other objects. e.g. bin, box, cabinet, folder, locker, safe, shelf, etc. Therefore a place is also a container but every container need not be a place.

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**Select Tangible things**

Take a “walk” through the system and select “tangible” things around you used in the problem domain. These may be characterized as all the remaining (not yet selected) “nouns” that make up the problem domain. Examples are: account, book, calendar, cash box, cash drawer, item, plan, procedure, product, schedule, skill, tool, etc

**Question No: 23 ( Marks: 3 )**

**What is the purpose of interaction diagram?**

**Answer: rep**

**Question No: 23 ( Marks: 2 )**

**Define cohesion.**

**Answer: (Page 72)**

Cohesion is an internal property of a module. Cohesion describes the intra-component linkages while couple shows the inter-component linkages. Cohesion measures the independence of a module.

**Question No: 23 ( Marks: 2 )**

**Keeping connie’s case study in mind, as discussed in lecture, list down whole parts structures which were identified.**

**Answer: (Page 100)**

Identify Whole-Part Structures

- ❖ A store as a whole is made up of cashiers, registers, and items.
- ❖ A register contains a cash drawer.
- ❖ A sale is constituted of sale line items.

**Question No: 23 ( Marks: -- )**

**Define data flow diagram**

**Answer: (Page 100)**

A data flow diagram (DFD) is a graphical representation of the "flow" of data through an information system, modeling its process aspects. Often they are a preliminary step used to create an overview of the system which can later be elaborated.

## **MIDTERM EXAMINATION - Spring 2010**

**Question No: 17 ( Marks: 2 )**

**Define abstraction?**

**Answer: rep**

**Question No: 18 ( Marks: 2 )**

**Is the design of software architecture a creative process?**

**Answer: rep**

**Question No: 19 ( Marks: 2 )**

**Suppose you are working as a software engineer involved in the development of an e-commerce website. What are the 2 most important characteristics your software must have?**

**Answer: rep**

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**Question No: 20 ( Marks: 3 )**

**What is the purpose of collaboration diagrams? Repeated**

**Answer: rep**

**Question No: 21 ( Marks: 3 )**

**What is the difference between Aggregation and Association? P5 chapter 7**

**Answer: rep**

**Question No: 22 ( Marks: 5 )**

**What parameters are used to measure and analyze design quality?**

**Answer: rep**

**Question No: 23 ( Marks: 5 )**

**How the objects are identified in Peter codd's technique? Repeated**

**Answer: rep**

**Question No: 24 ( Marks: 5 )**

**comparison of software architecture and Building architecture**

**Answer: (Page 115)**

When building a house, the architect, the general contractor, the electrician, the plumber, the interior designer, and the landscaper all have different views of the structure. Although these views are pictured differently, all are inherently related: together, they describe the building's architecture. The same is true with software architecture. Architectural design basically establishes the overall structure of a software system.

**Question No: 25 ( Marks: 10 )**

**Code example of High Coupling**

**Answer: [Click here for detail](#)**

**Tightly Coupled Example:**

```
public class CartEntry
{
    public float Price;
    public int Quantity;
}

public class CartContents
{
    public CartEntry[] items;
}

public class Order
{
    private CartContents cart;
    private float salesTax;

    public Order(CartContents cart, float salesTax)
    {
        this.cart = cart;
    }
}
```

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```

        this.salesTax = salesTax;
    }

    public float OrderTotal()
    {
        float cartTotal = 0;
        for (int i = 0; i < cart.items.Length; i++)
        {
            cartTotal += cart.items[i].Price * cart.items[i].Quantity;
        }
        cartTotal += cartTotal*salesTax;
        return cartTotal;
    }
}

```

## MIDTERM EXAMINATION

### Spring 2010

#### CS504- Software Engineering – I

**Question No: 16 ( Marks: 2 )**

Adding user interface detail in the SRS is controversial a creative process i.e by adding GUI detail to the SRS document, focus e shift from what to how (analysis design) do you agree ?

**Answer: (Page 115)**

Yes I am agree because Client appreciates more the contents of the SRS document if our SRS document contains the GUI details than if we don't have them there.

**Question No: 17 ( Marks: 2 )**

**Is the design software of architecture a creative process?**

**Answer: rep**

**Question No: 18 ( Marks: 3 )**

**What is the architecture design process explain briefly.**

**Answer: rep**

**Question No: 19 ( Marks: 3 )**

**What is textual analysis?**

**Answer: rep**

**Question No: 20 ( Marks: 5 )**

**What should be the consideration for the maintainable design?**

**Answer: rep**

**Question No: 21 ( Marks: 5 )**

**Discusses the relation between sequence diagram & logical complexity.**

**Answer: rep**

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Question No : 1 of 26

Marks: 1 (Budgeted Time 1 Min)

What is the most crucial non-functional requirement of a system to control radiation dosages that are emitted as treatment for cancer?

Answer ( Please select your correct option )

- ☒ Accuracy
- ☐ Security
- ☐ Reliability
- ☐ Usability

Correct answer solved by hadi  
Cell No:03228043306  
Email: usmanraj20@gmail.com

**Made By: Waqar Siddhu**

Question No : 2 of 26

Marks: 1 (Budgeted Time 1 Min)

Which of the following is NOT an objective for building an analysis model?

Answer ( Please select your correct option )

- ☐ Define set of software requirements
- ☐ Describe customer requirements
- ☒ Develop an abbreviated solution for the problem
- ☐ Establish basis for software design

**Made By: Waqar Siddhu**

Question No : 3 of 26

Marks: 1 (Budgeted Time 1 Min)

CRUD operations do not include \_\_\_\_\_

Answer ( Please select your correct option )

☐ Create

☐ Read

☐ Update

☒ Flow

Correct answer solved by hadi  
Cell No:03228043306  
Email: usmanraj20@gmail.com

**Made By: Waqar Siddhu**

Question No : 4 of 26

Marks: 1 (Budgeted Time 1 Min)

Which of the following is not considered as benefit of prototyping?

Answer ( Please select your correct option )

- ☐ Provide rapid feedback from the user to the designer
- ☐ Help validate requirements with fewer errors
- ☐ Enhance designer and user understanding of what system should accomplish
- ☐ Prototyping is a technique that can be used to reduce customer satisfaction

**Made By: Waqar Siddhu**

Question No : 5 of 26

Marks: 1 (Budgeted Time 1 Min)

'isAuthorized', 'assess Performance', are the examples of

Answer ( Please select your correct option )

☐ Properties

☒ Services

☐ Attributes

☐ Links

Correct answer solved by hadi  
Cell No:03228043306  
Email: usmanraj20@gmail.com

**Made By: Waqar Siddhu**

Question No : 6 of 26

Marks: 1 (Budgeted Time 1 Min)

Cohesion is about making sure that each component does \_\_\_\_\_ thing(s) and does it well

Answer ( Please select your correct option )

☐ Three

☐ Two

☐ Infinite

☒ One

**Made By: Waqar Siddhu**

Question No : 7 of 26

Marks: 1 (Budgeted Time 1 Min)

Static components of an OOA model are \_\_\_\_\_

Answer ( Please select your correct option )

☐

Focus on control

☐

Not reusable

☐

Sensitive to timing and event processing

☒

Structural in nature

Correct answer solved by hadi  
Cell No:03228043306  
Email: usmanraj20@gmail.com

**Made By: Waqar Siddhu**

Question No : 8 of 26

Marks: 1 (Budgeted Time 1 Min)

OOA is intended to define-----, their relationships, and their behavior.

Answer ( Please select your correct option )

☐

Variables

☐

Classes

☒

Objects

☐

Subjects

Correct answer solved by hadi  
Cell No:03228043306  
Email: usmanraj20@gmail.com

**Made By: Waqar Siddhu**

Question No : 9 of 26

Marks: 1 (Budgeted Time 1 Min)

Which of the following is not part of software architecture?

Answer ( Please select your correct option )

☒ Algorithm details

☐ Databases

☐ Data design

☐ Program structure

not include  
22 chp

**Made By: Waqar Siddhu**

Question No : 10 of 26

Marks: 1 (Budgeted Time 1 Min)

The vision statement should reflect a \_\_\_\_\_ that will satisfy the need of diverse customers.

Answer ( Please select your correct option )

☒

Balanced View

Correct answer solved by hadi  
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☐

System view

☐

Functional View

☐

Non Functional View

**Made By: Waqar Siddhu**

Question No : 11 of 26

Marks: 1 (Budgeted Time 1 Min)

Use cases describe the system as a \_\_\_\_\_ and hide the internal details from the user.

Answer ( Please select your correct option )

☒

Black Box

☐

White Box

☐

Key Box

☐

Inbox

**Made By: Waqar Siddhu**

Question No : 12 of 26

Marks: 1 (Budgeted Time 1 Min)

Software Engineering is the combination of tools, techniques and \_\_\_\_\_.

Answer ( Please select your correct option )

☐ Testing

☒ Processes

☐ Maintenance

☐ Design

Correct answer solved by hadi  
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**Made By: Waqar Siddhu**

Question No : 13 of 26

Marks: 1 (Budgeted Time 1 Min)

The project manager would need \_\_\_\_\_ to monitor and track the progress of the project.

Answer ( Please select your correct option )

☐ Feasibility report

☒ Requirement document

☐ Design document

☐ User manual

**Made By: Waqar Siddhu**

Question No : 14 of 26

Marks: 1 (Budgeted Time 1 Min)

A software requirement document describes all the \_\_\_\_\_ provided by the system along with the constraints under which it must operate.

Answer ( Please select your correct option )

☐ Conditions

☒ Services

☐ Tasks

☐ Actions

Correct answer solved by hadi  
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**Made By: Waqar Siddhu**

Question No : 15 of 26

Marks: 1 (Budgeted Time 1 Min)

Agreement, assignment, authorization, contract, delivery, deposit, incident etc are examples of \_\_\_\_\_.

Answer ( Please select your correct option )

☐ Object

☐ Participant

☐ Tangible

☒ Transaction

**Made By: Waqar Siddhu**

Question No : 16 of 26

Marks: 1 (Budgeted Time 1 Min)

In sequence diagram, a message is represented by a/an \_\_\_\_\_.

Answer ( Please select your correct option )

☒

Arrow

*A message is represented by an arrow between the life lines of two objects.*

☐

Straight line

☐

Dotted line

☐

Box

**Made By: Nagar Sidhu**

Question No : 17 of 26

Marks: 1 (Budgeted Time 1 Min)

\_\_\_\_\_ design decisions should be ones that remain constant in software development cycle.

Answer ( Please select your correct option )

☒ Early

Correct answer solved by hadi  
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Email: usmanraj20@gmail.com

☐ Late

☐ Heterogeneous

☐ Both a and b

**Made By: Waqar Siddhu**

Question No : 18 of 26

Marks: 1 (Budgeted Time 1 Min)

The data model consists of three pieces of interrelated information. Which of the following is not a piece of interrelated information?

Answer ( Please select your correct option )

☐

Attributes

☐

Data objects

☐

Relationships

☒

Controls

**Made By: Waqar Siddhu**

Question No : 19 of 26

Marks: 1 (Budgeted Time 1 Min)

The first most important step in developing any system is to identify \_\_\_\_\_.

Answer ( Please select your correct option )

☐

Purpose of the system

☐

Implementation of the system

☐

Design of the system

☒

Architecture of the system

**Made By: Waqar Siddhu**

Question No : 20 of 26

Marks: 1 (Budgeted Time 1 Min)

When measure of independence of a module or component is low to the other, changes in one component have effect on other component is \_\_\_\_\_.

Answer ( Please select your correct option )

☐ No.

☐ Low

☒ High

☐ Equal

Correct answer solved by hadi  
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Email: usmanraj20@gmail.com

**Made By: Waqar Siddhu**

Question No : 21 of 26

Marks: 2 (Budgeted Time 4 Min)

Which of the following statement(s) is/are true about software design?

- (a) Common coupling occurs when some global data repository is shared by two or more modules.
- (b) Polymorphism is a design principle that helps the code to be flexible and reused.
- (c) Software reuse injects more reliability to code.

Answer ( [Please click here to Add Answer](#) )



**Made By: Waqar Siddhu**

Question No : 22 of 26

Marks: 2 (Budgeted Time 4 Min)

As per Peter Coad methodology, identify the transaction object(s) from the list below.

- 1) Sale
- 2) Purchase
- 3) Computer
- 4) Cashier

Answer ( [Please click here to Add Answer](#) )



Correct answer solved by hadi  
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Email: usmanraj20@gmail.com

**Made By: Waqar Siddhu**

Question No : 23 of 26

Marks: 3 (Budgeted Time 6 Min)

What is the difference between Aggregation and Association?

Answer ( [Please click here to Add Answer](#) )

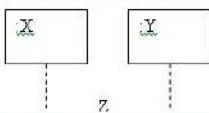


**Made By: Waqar Siddhu**

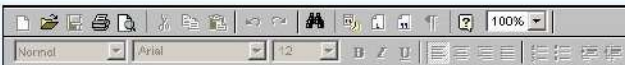
Question No : 24 of 26

Marks: 3 (Budgeted Time 6 Min)

Consider the following diagram:



Answer ( [Please click here to Add Answer](#) )



**Made By: Waqar Siddhu**

Question No : 25 of 26

Marks: 5 (Budgeted Time 10 Min)

A client sends an authentication request (with username and password) to server, server verifies the request and returns a response to the client.

Draw a Sequence level diagram of the above scenario.

Answer ( [Please click here to Add Answer](#) )



**Made By: Waqar Siddhu**

Question No : 26 of 26

Marks: 5 (Budgeted Time 10 Min)

Apply Cohesion on the following code.

Class Reserv

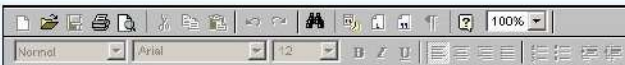
{

private:

int roomno;

string type;

Answer ( [Please click here to Add Answer](#) )



**Made By: Waqar Siddhu**



## CS504-Software Engineering -1

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### CS504-MIDTERM SOLVED MCQS WITH REFERENCES



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## MIDTERM EXAMINATION 2010

**Question No: 1 ( Marks: 1 ) - Please choose one**

**By following modern system engineering practices simulation of reactive systems is no longer necessary**

- ▶ True
- ▶ **False** [click here for detail](#)

**Question No: 2 ( Marks: 1 ) - Please choose one**

**The state transition diagram**

- ▶ depicts relationships between data objects
- ▶ depicts functions that transform the data flow
- ▶ indicates how data are transformed by the system
- ▶ **indicates system reactions to external events** [click here for detail](#)

**Question No: 3 ( Marks: 1 ) - Please choose one**

**Control flow diagrams are**

- ▶ **needed to model event driven systems.** [Click here for detail](#)
- ▶ required for all systems
- ▶ used in place of data flow diagrams
- ▶ useful for modeling user interfaces

**Question No: 4 ( Marks: 1 ) - Please choose one**

**A change becomes ----- because of close presence of data and functions**

- ▶ **Localized** (page 81)
- ▶ Private
- ▶ Global
- ▶ Accessible

**Question No: 5 ( Marks: 1 ) - Please choose one**

**A complex System evolves from a**

- ▶ **smaller system** page 83 [click here for detail](#)
- ▶ medium sysetem
- ▶ bigger system
- ▶ non of the given

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**Question No: 6 ( Marks: 1 ) - Please choose one**

**Coupling is a measure of -----of a module or component.**

▶ **Independent** (Page 71)

- ▶ Dependent
- ▶ Closeness
- ▶ All of the given

**Question No: 7 ( Marks: 1 ) - Please choose one**

**Software Maintenance phase involves**

- ▶ Debugging
- ▶ Adding new features
- ▶ making changes
- ▶ **all of the given** [click here for detail](#)

**Question No: 8 ( Marks: 1 ) - Please choose one**

**The hardest single part of building a software system is deciding precisely ----- to build.**

▶ **what** (Page 17)

- ▶ How
- ▶ When
- ▶ Why

**Question No: 9 ( Marks: 1 ) - Please choose one**

**Interaction Diagrams depict the ----- Behavior of the system**

- ▶ Static
- ▶ **Dynamic** (Page 106)
- ▶ Active
- ▶ Non of the above

**Question No: 10 ( Marks: 1 ) - Please choose one**

**In Sequence Diagrams the time required by the receiver object to process the message is denoted by an-----**

▶ **Activation Box** (Page 108)

- ▶ Simple Box
- ▶ Arrow
- ▶ Non of the above

**Question No: 11 ( Marks: 1 ) - Please choose one**

**A poorly designed interface can cause a user to make catastrophic errors is one of the motivations for GUI.**

▶ **True** (Page 62)

- ▶ False

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**Question No: 12 ( Marks: 1 ) - Please choose one**  
**Identifying System Features include -----**

- ▶ log important information
- ▶ conduct business
- ▶ analyze business results
- ▶ **all of the above** (Page 98)

**Question No: 13 ( Marks: 1 ) - Please choose one**  
**Establishing responsibilities for objects includes**

- ▶ Generalization Relationships
- ▶ Specialization Relationships
- ▶ **all of the above** (Page 86)
- ▶ identifying Association relationships

**Question No: 14 ( Marks: 1 ) - Please choose one**  
**Which of the following is a fact finding method?**

- ▶ Site visits
- ▶ **Prototyping**
- ▶ Study of similar systems
- ▶ All of given

Seven common fact-finding methods (Sampling, Research, Observation, Questionnaires, Interviews, Prototyping, Joint Requirements Planning) are introduced as a means to discover requirements.

**Question No: 15 ( Marks: 1 ) - Please choose one**

**Windows mobile is a popular mobile operating system which seen commonly on PDAs. Which of the following category pair(s) best describe(s) it?**

- ▶ **Application software, embedded software**
- ▶ system software, web-based software
- ▶ application software, scientific software
- ▶ system software, embedded software

**Question No: 16 ( Marks: 1 ) - Please choose one**  
**The Object Oriented approach**

- ▶ **improves the reusability of codes.** (Page 83)
- ▶ makes objects less independent.
- ▶ increases testing time.
- ▶ complicates mapping of a system model to an implementation model.

The object-oriented approach improves the maintainability, reusability, and extensibility of the software [Click here for Detail](#)

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## MIDTERM EXAMINATION 2010

**Question No: 1 ( Marks: 1 ) - Please choose one**

**Most software continues to be custom built because**

- ▶ Software is easier to build without using someone else's components.
- ▶ **Off the shelf software components are not commonly available** [Click here for detail](#)
- ▶ Component reuse is common in the software world
- ▶ Reusable components are too expensive to use

**Question No: 2 ( Marks: 1 ) - Please choose one**

**Which of the items listed below is not one of the software engineering layers?**

- ▶ Tools
- ▶ Process
- ▶ **Manufacturing** [Click here for detail](#)
- ▶ Methods

**Question No: 3 ( Marks: 1 ) - Please choose one**

**Which of these people would not be likely to part of the FAST team?**

- ▶ hardware and software engineers
- ▶ manufacturing representative
- ▶ marketing representatives
- ▶ **senior financial officers** **APRACTITIONER'S APPROACH (Page 305)**

**Question No: 4 ( Marks: 1 ) - Please choose one**

**An state transition can only occur when triggered by a(n)\_\_\_\_\_**

- ▶ actor
- ▶ collaboration attempt
- ▶ **event**
- ▶ none of the given

**The evolution of the system depends on a sequence of state transitions triggered by a set of coming events.**

**Question No: 5 ( Marks: 1 ) - Please choose one**

**A change becomes ----- because of close presence of data and functions**

- ▶ **Localized (page 81)**
- ▶ Private
- ▶ Global
- ▶ Accessible

**Question No: 6 ( Marks: 1 ) - Please choose one**

**A process which does not take input is called**

- ▶ **miracle process**      Miracle": Process has output, but no input      [Click here for detail](#)
- ▶ core process
- ▶ secondary process
- ▶ all of the above

**Question No: 7 ( Marks: 1 ) - Please choose one**

**Data Store notation in DFDs presents.**

- ▶ Data input
- ▶ Data output
- ▶ **Data input and Data output**      (Page 54)
- ▶ non of the given

**Question No: 8 ( Marks: 1 ) - Please choose one**

**Coupling is a measure of -----of a module or component.**

- ▶ **Independent**      (Page 71)
- ▶ Dependent
- ▶ Closeness
- ▶ All of the given

**Question No: 9 ( Marks: 1 ) - Please choose one**

**If a problem is identified and fixed at a -----stage in the software development process, it will cost much more than if it was fixed at and -----stage.**

- ▶ eralier & lator
- ▶ after & before
- ▶ none of the above
- ▶ **later and earlier**      (Page 17)

**Question No: 10 ( Marks: 1 ) - Please choose one**

**Once these requirements have been finalized, the ----- process starts.**

- ▶ **Construction**      (Page 18)
- ▶ Development
- ▶ Maintenance phase
- ▶ Construction and Development

**Question No: 11 ( Marks: 1 ) - Please choose one**

**Identifying Whole-Part structures (Aggregations) means what are my -----**

- ▶ **Components**      (Page 95)
- ▶ Structures
- ▶ Objec
- ▶ Classes

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**Question No: 12 ( Marks: 1 ) - Please choose one**

**Data design actually begins during the creation of the analysis model, not the architectural model.**

- ▶ **True** [Click here for detail](#)
- ▶ False

**Question No: 13 ( Marks: 1 ) - Please choose one**

**System Architecture is important to consider because it helps in making -----**

- ▶ Mutual communication.
- ▶ Early design decisions.
- ▶ Reusable abstraction of a system.
- ▶ **All of the above** (Page 118)

**Question No: 14 ( Marks: 1 ) - Please choose one**

**Software architecture is "the ----- of the components of a program/system, their interrelationships, and principles and guidelines governing their design and evolution over time."**

- ▶ combination
- ▶ collection
- ▶ **structure** (Page 117)
- ▶ unification

**Question No: 15 ( Marks: 1 ) - Please choose one**

**Cohesion is about making sure each component does \_\_\_\_\_ thing and does it well.**

- ▶ three
- ▶ two
- ▶ infinite
- ▶ **one** [Click here for detail](#)

**Question No: 16 ( Marks: 1 ) - Please choose one**

**Following are some statements associated with data flow diagrams. Identify the correct statement from among them.**

- ▶ DFDs are used to model complex interfaces.
- ▶ **DFDs are used to represent only functional processing, data stores and data movements between functions.** DFD over flow charts(Page 2)
- ▶ DFDs depict only processes which can be decomposed.
- ▶ DFDs do not show external data sources and external data sinks

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## MIDTERM EXAMINATION 2010

**Question No: 1 ( Marks: 1 ) - Please choose one**

**UML (unified modeling language) analysis modeling focuses on the \_\_\_\_\_ .**

- ▶ behavioral model and environment model.
- ▶ behavioral model and implementation model.
- ▶ user model and environmental model
- ▶ **user model and structural model**      APRACTITIONER'S APPROACH (Page 604)

**Question No: 2 ( Marks: 1 ) - Please choose one**

**The data flow diagram must be augmented by descriptive text in order to describe the functional requirements for a software product.**

- ▶ **True**      [Click here for detail](#)
- ▶ False

**Question No: 3 ( Marks: 1 ) - Please choose one**

**It is not possible to use ordinary data flow diagrams to model the functional requirements of real-time systems.**

- ▶ True
- ▶ **False**      [Click here for detail](#)

**Question No: 4 ( Marks: 1 ) - Please choose one**

**An object or class may further be classified on the basis of**

- ▶ Behavior Driven attributes
- ▶ Data Driven attributes
- ▶ Responsibility Driven attributes
- ▶ **All of the above**      (Page 85)

**Question No: 5 ( Marks: 1 ) - Please choose one**

**A public Interface provides a way for \_\_\_\_\_ with other Classes.**

- ▶ Communication
- ▶ **Accessibility**      [Click here for detail](#)
- ▶ Reaching
- ▶ All of the above

**Question No: 6 ( Marks: 1 ) - Please choose one**

**DFD Notation contains**

- ▶ Data Store
- ▶ Extenal Agents
- ▶ Processes
- ▶ **All of the given**      (Page 51)

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**Question No: 7 ( Marks: 1 ) - Please choose one**

**A complex System evolves from a**

- ▶ **smaller system** (Page 69)
- ▶ medium sysetem
- ▶ bigger system
- ▶ non of the given

**Question No: 8 ( Marks: 1 ) - Please choose one**

**Requirement engineering mainly deals with the ----- of the system**

- ▶ **definition phase** (Page 16)
- ▶ development phase
- ▶ maintenance
- ▶ non of the above

**Question No: 9 ( Marks: 1 ) - Please choose one**

**Insufficient user involvement leads to ----- products.**

- ▶ **unacceptable** (Page 19)
- ▶ ambiguous
- ▶ dissatisfaction of customer to
- ▶ non of the above

**Question No: 10 ( Marks: 1 ) - Please choose one**

**Which one is not the purpose of Interaction Diagrams ?**

- ▶ Model interactions between objects
- ▶ Assist in understanding how a system (a use case) actually works
- ▶ Identify responsibilities/operations and assign them to classes
- ▶ **Identify dependencies among objecs** (Page 106)

**Question No: 11 ( Marks: 1 ) - Please choose one**

**Collaboration diagrams have basically two types of components: objects and -----**

- ▶ **messages** (Page 111)
- ▶ methods
- ▶ classes
- ▶ non of the above

**Question No: 12 ( Marks: 1 ) - Please choose one**

**The scope description establishes the -----between the system we are developing and everything else in the universe**

- ▶ **Boundary** (Page 31)
- ▶ Balance
- ▶ Constraint
- ▶ None of the given

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**Question No: 13 ( Marks: 1 ) - Please choose one**

**Which one is not a type of messages which Sequence Diagrams Depict**

- ▶ Synchronous
- ▶ Asynchronous
- ▶ Create
- ▶ **Update (Page 108)**

**Question No: 14 ( Marks: 1 ) - Please choose one**

**What is the most crucial non-functional requirement of a system to control radiation dosages that are emitted as treatment for cancer?**

- ▶ Security
- ▶ Reliability
- ▶ Usability
- ▶ **Accuracy (not sure)**

**Question No: 15 ( Marks: 1 ) - Please choose one**

**Given below are some statements associated with data flow diagrams. Identify the correct statement from among them.**

- ▶ Data flow is made use of to model what systems do.
- ▶ Flows of data can take place from a process to a sink.
- ▶ All processes have to be levelled or decomposed.
- ▶ **Context diagram shows the major system processes. (Page 54)**

**Question No: 16 ( Marks: 1 ) - Please choose one**

**What would be the most suitable architecture to develop a commercial web page to do business transactions over the internet?**

- ▶ **Client server model (p129)**
- ▶ Island model
- ▶ RAD model
- ▶ Repository model

## **MIDTERM EXAMINATION 2010**

**Question No: 1 ( Marks: 1 ) - Please choose one**

**The first step in any OOA process model is to\_\_\_\_\_**

- ▶ **build an object-relationship model (p90)**
- ▶ define collaborations between objects
- ▶ elicit customer requirements
- ▶ select a representation language

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**Question No: 2 ( Marks: 1 ) - Please choose one**

**Dynamic components of an OOA model are \_\_\_\_\_**

- ▶ not reusable
- ▶ **sensitive to timing and event processing**
- ▶ stable throughout the operational life of an application
- ▶ structural in nature

Dynamic components:

Focus on control and are sensitive to timing and event processing.

Dynamic view of Communication,

Dynamic view of Control and Time.

**Question No: 3 ( Marks: 1 ) - Please choose one**

**Events occur whenever a(n)\_\_\_\_\_**

- ▶ **actor and the OO system exchange information** APRACTITIONER'S APPROACH (Page 622)
- ▶ class operation is invoked
- ▶ messages are passed between objects
- ▶ all of the given

**Question No: 4 ( Marks: 1 ) - Please choose one**

**By following modern system engineering practices simulation of reactive systems is no longer necessary**

- ▶ True
- ▶ **False** [Click here for detail](#)

**Question No: 5 ( Marks: 1 ) - Please choose one**

**Information flow represents the manner in which data and control**

- ▶ are related to one another.
- ▶ **changes as each moves through the system.** APRACTITIONER'S APPROACH (Page 312)
- ▶ will be implemented in the final design.
- ▶ none of the given

**Question No: 6 ( Marks: 1 ) - Please choose one**

**Which view should be consider first during software requirements analysis?**

- ▶ actor view
- ▶ data view
- ▶ essential view
- ▶ **implementation view** APRACTITIONER'S APPROACH (Page 316)

**Question No: 7 ( Marks: 1 ) - Please choose one**

**State Transition Diagram is helpful in determining**

- ▶ **Business Understanding (Page 52)**
- ▶ Process Flow
- ▶ Data store
- ▶ Non of the given

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**Question No: 8 ( Marks: 1 ) - Please choose one**

**A better Design has an objective achieve**

- ▶ High Cohesion
- ▶ Low Cohesion
- ▶ Low Coupling
- ▶ **High cohesion and Low coupling**      **APRACTITIONER'S APPROACH** (Page 316)

**Question No: 9 ( Marks: 1 ) - Please choose one**

**The process of defining attributes is called**

- ▶ **What I know**      (Page 95)
- ▶ Who know me
- ▶ Whom I know
- ▶ Non of the above

**Question No: 10 ( Marks: 1 ) - Please choose one**

**Which of the following is not an objective for building an analysis model?**

- ▶ define set of software requirements
- ▶ describe customer requirements
- ▶ **develop an abbreviated solution for the problem**      **APRACTITIONER'S APPROACH** (Page 329)
- ▶ establish basis for software design

**Question No: 11 ( Marks: 1 ) - Please choose one**

**The output of this design process is a description of the**

- ▶ **Software Architecture**      (Page 115)
- ▶ Software Code
- ▶ Software
- ▶ Non of the above

**Question No: 12 ( Marks: 1 ) - Please choose one**

**Which of these characteristics are true of a data warehouse, but not a typical database?**

- ▶ business level orientation and large size
- ▶ **currency and correctness of information**      [Click here for detail](#)
- ▶ integration and non volatility
- ▶ all of the given

**Question No: 13 ( Marks: 1 ) - Please choose one**

**Identifying System Features include -----**

- ▶ log important information
- ▶ conduct business
- ▶ analyze business results
- ▶ **all of the above**      (Page 98)

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**Question No: 14 ( Marks: 1 ) - Please choose one**

**The two main documents produced during this phase are Requirement Statement and Requirement Specification. They are also called Requirement Definition and -----**

- ▶ **Functional Specification (Page 25)**
- ▶ Mathematical specification
- ▶ System Specification
- ▶ None of the given

**Question No: 15 ( Marks: 1 ) - Please choose one**

**Project ----- defines the concept and range of the proposed solution, and limitations identify certain capabilities that the product will not include**

- ▶ **Scope (Page 30)**
- ▶ Agreement
- ▶ Plan
- ▶ None of the given

**Question No: 16 ( Marks: 1 ) - Please choose one**

**Windows mobile is a popular mobile operating system which seen commonly on PDAs. Which of the following category pair(s) best describe(s) it?**

- ▶ **application software, embedded software (not sure)**
- ▶ system software, web-based software
- ▶ application software, scientific software
- ▶ system software, embedded software

## **MIDTERM EXAMINATION 2010**

**Question No: 1 ( Marks: 1 ) - Please choose one**

**Control flow diagrams are.**

- ▶ **Needed to model event driven systems. [Click here for detail](#)**
- ▶ Required for all systems.
- ▶ Used in place of data flow diagrams.
- ▶ Useful for modeling user interfaces.

**Question No: 2( Marks: 1 ) - Please choose one**

**The Data dictionary contain description of each software.**

- ▶ Configuration item.
- ▶ **Data object APRACTITIONER'S APPROACH (Page 356)**
- ▶ Diagram
- ▶ Notation

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**Question No: 3( Marks: 1 ) - Please choose one**

**A cohesion class is one which emphasize -----unit of----**

- ▶ Single and multiple.
- ▶ Multiple and functionality
- ▶ Functional and single
- ▶ **Single and functional (Page 72)**

**Question No: 4( Marks: 1 ) - Please choose one**

**Flow charts represent.**

- ▶ **Sequence. (Page 50)**
- ▶ Random
- ▶ Parallel
- ▶ Non of above

**Question No: 5( Marks: 1 ) - Please choose one**

**----- gives OO the ability to handle essential**

- ▶ **Decentralization (Page 80 )**
- ▶ Centralization
- ▶ Decentralization and Centralization
- ▶ Non of above

**Question No: 6( Marks: 1 ) - Please choose one**

**In sequence Diagram events are organized in a-----time line**

- ▶ **vertical (Page 106)**
- ▶ horizontal
- ▶ Vertical and Horizontal
- ▶ Non of above

**Question No: 7( Marks: 1 ) - Please choose one**

**Asynchronous messages are denoted**

- ▶ **Half Arrow (Page 109)**
- ▶ Simple Line
- ▶ Full Arrow
- ▶ Non of above

**Question No: 8( Marks: 1 ) - Please choose one**

**Software crisis came in 1960 what is the main reason to for the crisis**

- ▶ **Software development technique (Page 4)**
- ▶ Hardware

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- ▶ Software
- ▶ Non of above

**Question No: 10 ( Marks: 1 ) - Please choose one**

**central role of the software requirement Document in entire development process is related to**

- ▶ Project planning
- ▶ Project tracing
- ▶ Change control
- ▶ **All of above (Page 19)**

**Question No: 11 ( Marks: 1 ) - Please choose one**

**The scope description establishes the ----- between the system we are developing and everything else in the universe. Project planning**

- ▶ **Boundary (Page 31)**
- ▶ System
- ▶ limitation
- ▶ None of above

**Question No: 12 ( Marks: 1 ) - Please choose one**

**which one is not the type of message which sequence diagram depict**

- ▶ Synchronous
- ▶ asynchronous
- ▶ Create
- ▶ **Update (Page 108)**

**Question No: 13 ( Marks: 1 ) - Please choose one**

**cohesion is about making sure each component does----- thing and does it well**

- ▶ Three
- ▶ infinite
- ▶ **one APRACTITIONER'S APPROACH (Page 381)**
- ▶ two

**Question No: 14 ( Marks: 1 ) - Please choose one**

**what is the most important software quality of a web based transaction**

- ▶ efficiency
- ▶ Modularity
- ▶ **usability [click here for detail](#)**
- ▶ Security

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Question No : 1 of 26

Marks: 1 (Budgeted Time 1 Min)

The state transition diagram \_\_\_\_\_

Answer ( Please select your correct option )

- ☐ Depicts relationships between data objects
- ☐ Depicts functions that transform the data flow
- ☐ Indicates how data are transformed by the system
- ☒ Indicates system reactions to external events

Correct answer solved by hadi  
Cell No:03228043306  
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**Made By: Waqar Siddhu**

Question No : 2 of 26

Marks: 1 (Budgeted Time 1 Min)

Domain model is \_\_\_\_\_ step in making an efficient system.

Answer ( Please select your correct option )



Initial



Secondary



Last



Middle

**Made By: Waqar Siddhu**

Question No : 3 of 26

Marks: 1 (Budgeted Time 1 Min)

CRUD operations do not include \_\_\_\_\_

Answer ( Please select your correct option )

☐ Create

☐ Read

☐ Update

☒ Flow

Correct answer solved by hadi  
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Question No : 4 of 26

Marks: 1 (Budgeted Time 1 Min)

In a \_\_\_\_\_, each program module is represented by a rectangular box.

Answer ( Please select your correct option )



Use case diagram



Document flow diagram



Class diagram



Data flow diagram

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Question No : 5 of 26

Marks: 1 (Budgeted Time 1 Min)

'isAuthorized', 'assess Performance', are the examples of

Answer ( Please select your correct option )

☐ Properties

☒ Services

☐ Attributes

☐ Links

Correct answer solved by hadi  
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**Made By: Waqar Siddhu**

Question No : 6 of 26

Marks: 1 (Budgeted Time 1 Min)

What would be the most suitable architecture to develop a commercial web page to do business transactions over the internet?

Answer ( Please select your correct option )

☒

Client server model

☐

Island model

☐

RAD model

☐

Repository model

**Made By: Waqar Siddhu**

Question No : 7 of 26

Marks: 1 (Budgeted Time 1 Min)

UML (unified modeling language) analysis modeling focuses on the \_\_\_\_\_.

Answer ( Please select your correct option )

☐ Behavioral model and environment model

☐ Behavioral model and implementation model

☐ User model and environmental model

☒ User model and structural model

Correct answer solved by hadi  
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**Made By: Waqar Siddhu**

Question No : 8 of 26

Marks: 1 (Budgeted Time 1 Min)

The vertical dotted lines in sequence diagram represent \_\_\_\_\_

Answer ( Please select your correct option )

☐ Creation of objects

☒ Life line of objects

☐ Structure of objects

☐ Services of objects

**Made By: Waqar Siddhu**

Question No : 9 of 26

Marks: 1 (Budgeted Time 1 Min)

40-60% of all defects found in software projects can be traced back to poor \_\_\_\_\_

Answer ( Please select your correct option )

☒ Requirements

Correct answer solved by hadi  
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☐ Design

☐ Coding

☐ Testing

**Made By: Waqar Siddhu**

Question No : 10 of 26

Marks: 1 (Budgeted Time 1 Min)

A SRS is said to be \_\_\_\_\_ if and only if, every requirement stated therein has only one interpretation.

Answer ( Please select your correct option )

☐ Compact

☒ Unambiguous

☐ Consistent

☐ Detailed

**Made By: Waqar Siddhu**

Question No : 11 of 26

Marks: 1 (Budgeted Time 1 Min)

Which of the following is NOT among directly related construction activities of software development?

Answer ( Please select your correct option )

☒ Management

Correct answer solved by hadi  
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☐ Testing

☐ Requirements Elicitation

☐ Design development

**Made By: Waqar Siddhu**

Question No : 12 of 26

Marks: 1 (Budgeted Time 1 Min)

The \_\_\_\_\_ relationship is kind of a generalization-specialization relationship.

Answer ( Please select your correct option )

☐ Composition

☐ Uses

☐ Association

☒ Extends

**Made By: Waqar Siddhu**

Question No : 13 of 26

Marks: 1 (Budgeted Time 1 Min)

\_\_\_\_\_ relationship is concerned with classes not with the class instances.

Answer ( Please select your correct option )

☐ Aggregation

☒ Inheritance

☐ Composition

☐ Association

Correct answer solved by hadi  
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**Made By: Waqar Siddhu**

Question No : 14 of 26

Marks: 1 (Budgeted Time 1 Min)



\_\_\_\_\_ activities are directly related to software development.

Answer ( Please select your correct option )

☒

Construction

☐

Management

☐

Quality assurance

☐

Project planning

**Made By: Waqar Siddhu**

Question No : 15 of 26

Marks: 1 (Budgeted Time 1 Min)

Two components of use-case model are actors and \_\_\_\_\_.

Answer ( Please select your correct option )

☐ Functional requirements

☐ Processes

☒ Use-cases

Correct answer solved by hadi  
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☐ System boundary

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Question No : 16 of 26

Marks: 1 (Budgeted Time 1 Min)



\_\_\_\_\_ is not the part of Peter Coad methodology.

Answer ( Please select your correct option )

☐ Select actors

☐ Select participants

☐ Select places

*4. Select Transactions*

*5. Select Container Objects*

*6. Select Tangible things*

☐ Select actions



**Made By: Waqar Siddhu**

Question No : 17 of 26

Marks: 1 (Budgeted Time 1 Min)

Transactions are the \_\_\_\_\_ that must be remembered through time.

Answer ( Please select your correct option )



Events

*pg: 93*

Correct answer solved by hadi  
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Actions



Triggers



Methods

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Question No : 16 of 26

Marks: 1 (Budgeted Time 1 Min)

The verbs in your problem domain usually indicate some of the \_\_\_\_\_ required of the associated object.

Answer ( Please select your correct option )

☒ Services

☐ Objects

☐ Collaborations

☐ Actors

**Made By: Waqar Siddhu**

Question No : 19 of 26

Marks: 1 (Budgeted Time 1 Min)

For purpose of behavior modeling, a state is any \_\_\_\_\_.

Answer ( Please select your correct option )

☐ Consumer or producer of data

☐ Data object hierarchy

☒ Observable mode of behavior

☐ Well defined process

**Made By: Waqar Siddhu**

Question No : 20 of 26

Marks: 1 (Budgeted Time 1 Min)

A complex System evolves from a \_\_\_\_\_

Answer ( Please select your correct option )

☐ Smaller system

☒ Simpler system

☐ Bigger system

☐ Medium system

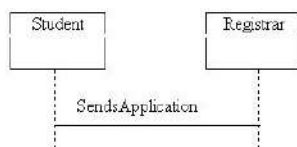
Correct answer solved by hadi  
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Email: usmanraj20@gmail.com

**Made By: Waqar Siddhu**

Question No : 21 of 26

Marks: 2 (Budgeted Time 4 Min)

Identify the object(s) from the following sequence diagram:



Answer ( Please [click here](#) to Add Answer )

Normal Arial 12 B I U

*Answer: Object: Student and Application.*

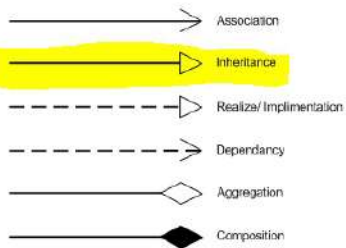
**Made By: Waqar Siddhu**

Question No : 22 of 26

Marks: 2 (Budgeted Time 4 Min)

As a software designer, you are assigned to prepare the Class diagram for the object Model. What is the notation for showing Inheritance relationship?

Answer



**Made By: Waqar Siddhu**

Question No : 23 of 26

Marks: 3 (Budgeted Time 6 Min)

Define following:-

- 1) Synchronous Messages
- 2) Asynchronous messages

Answer ( [Please click here to Add Answer](#) )



*Asynchronous messages are "signals," denoted by a half arrow. They do not block the caller. Asynchronous messages typically perform the following actions:*

- ❖ *Create a new thread*
- ❖ *Create a new object*
- Communicate with a thread that is already running*

*Synchronous Messages:-*

*Synchronous messages are "call events" and are denoted by the full arrow. They represent nested flow of control which is typically implemented as an operation call. In case of a synchronous message, the caller waits for the called routine to complete its operation before moving forward.*

**Made By: Waqar Siddhu**

Question No : 24 of 26

Marks: 3 (Budgeted Time 6 Min)

Ali is considered as an object. What are the characteristics which decide Ali is an Object?

Answer ( [Please click here to Add Answer](#) )

Normal Arial 12 B I U

*Answer: Ali could be a person, a student, or employee.  
he will have characteristic which tell his class for which class of object it is.*

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Question No : 25 of 26

Marks: 5 (Budgeted Time 10 Min)

Consider the following five code fragments.

| 1                                                                                                                                 | 2                                                                                                                                                                   | 3                                                                                                                                                          |
|-----------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>class File {<br/>void OpenFile () { ... }<br/>void SearchIndex () { ... }<br/>void CloseFile () { ... }<br/>}</pre>          | <pre>class AreaFun {<br/>double CircleArea()<br/>{ ... }<br/>double RectangleArea() { ... }<br/>}<br/>double TriangleArea() { ... }<br/>}</pre>                     | <pre>class MyClass {<br/>void OpenFile () { ... }<br/>void SearchIndex () { ... }<br/>void InitPrinter () { ... }<br/>void Print index { ... }<br/>}</pre> |
| 4                                                                                                                                 | 5                                                                                                                                                                   |                                                                                                                                                            |
| <pre>class POST {<br/>void CheckPower () { ... }<br/>void InitializeDisc () { ... }<br/>void LoadBootstrap () { ... }<br/>}</pre> | <pre>Class MakeSoup {<br/>void BoilWater () { ... }<br/>Void AddIngredients () { ... }<br/>void MixIngredients () { ... }<br/>void ServeSoup () { ... }<br/>}</pre> |                                                                                                                                                            |

answer: 3

Which of the above code fragments show the weakest level of cohesion? Brief it with reason.

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Question No : 26 of 26

Marks: 5 (Budgeted Time 10 Min)

Suppose you were analyzing a Requirement statement, How will you determine that an object belongs to a certain class?

Answer ( [Please click here to Add Answer](#) )



*analyzed on the basis of their characteristics , behaviour and states. things with same (all these) will be in same class. if we need many items of same (all these things) we will define a class for them.*

Correct answer solved by hadi  
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Rubaisha Skype Group  
**Lec#01 SUBJECTIVE**

**Q : What is software? (marks 2)**

**Ans.** When we write a program for computer we named it as software. But software is not just a program; many things other than the program are also included in software.

**Q : How many items in software? Write name and explain . (marks 3)**

**Ans.** ☐ **Program:** The program or code itself is definitely included in the software.

☐ **Data:** The data on which the program operates is also considered as part of the software.

☐ **Documentation:** Another very important thing that most of us forget is documentation. All the documents related to the software are also considered as part of the software.

**Q : What is Engineering? (marks 2)**

**Ans.** The process of productive use of scientific knowledge is called engineering.

**Q : Difference between computer science and software engineering. (marks 5)**

**Ans.** When we use physics in making machines like engines or cars then it is called mechanical engineering. And when we apply the knowledge of physics in

developing electronic devices then the process is called electrical engineering. The relation of computer science with software engineering is similar as the relation of physics with the electrical, mechanical or civil engineering or for that matter the relation of any basic science with any engineering field. So, This is the process of utilizing our knowledge of computer science in effective production of software systems.

**Q : Difference between Software and Other Systems. (marks 3)**

**Ans.** If the software has a bug and that bug was present in the older CD then that will remain in the new one. This is a fundamental difference between software and other systems.

**Q : What is Software Crisis? (marks 3)**

**Ans.** Software Crisis used to process the data of census. More powerful hardware resulted into the development of more powerful and complex software. Those very complex software was very difficult to write. So the tools and techniques that were used for less complex software became inapplicable for the more complex software.

**Q : What is software engineering as by IEEE? (marks 2)**

**Ans.** The application of a systematic, disciplined, quantifiable approach to the development, operation, and maintenance of software; that is, the application of engineering to software. And the IEEE stands for International institute of Electric and Electronic Engineering.

**Q : What is software engineering used in software production? (marks 5)**

**Ans.** Software

Engineering is the combination of all the tools, techniques, and processes that used in software production.

- Programming Language
- Programming Language Design
- Software Design Techniques
- Tools
- Testing
- Maintenance
- Development

**Q : Characteristics of well-Engineered software? (marks 3)**

**Ans.**

- It is reliable
- It has good user-interface
- It has acceptable performance
- It is of good quality
- It is cost-effective

**Q : What is Law of balancing act in software? (marks 5)**

**Ans.**

Software Engineering is actually the balancing act. You have to balance many things like cost, user friendliness, Efficiency, Reliability etc. You have to analyze which one is the more important feature for your software is it reliability, efficiency, user friendliness or something

else. There is always a trade-off among all these requirements of software. It may be the case that if you try to make it more user-friendly then the efficiency may suffer. And if you try to make it more cost-effective then reliability may suffer. Therefore there is always a trade-off between these characteristics of software. These requirements may be conflicting

## **Lec#02 SUBJECTIVE**

**Q :.What is the Construction and major types of Construction? (marks 5)**

**Ans.** The construction activities are those that directly related to the development of software, e.g. gathering the requirements of the software, develop design, implement and test the software etc. Some of the major construction activities are listed below.

- Requirement Gathering
- Design Development
- Coding
- Testing

**Q : Write the name of the major activities of management? (marks 3)**

**Ans.**

- Project Planning and Management
- Configuration Management
- Software Quality Assurance
- Installation and Training

**Q : Write the name of the major stage of software development loop? (marks 2)**

**Ans.**

1. Problem Definition
2. Technical Development
3. Solution Integration
4. Status Quo

**Q : How many software Engineering phase? and write the name. (marks 3)**

**Ans.** There are four basic phase of S.E

- 1.Vision
- 2.Definition
- 3.Development
- 4.Maintenance

### **Lec#03 SUBJECTIVE**

**Q : Write the name of the role of software Requirement. (marks 3)**

**Ans.**

- 1.Project Planning
- 2.project Tracking
- 3.Change Control

4. System Testing

5. User Documentation

6. Construction Process

### **Lec#04 SUBJECTIVE**

**Q : Write the name of the Requirement Statement Characteristics. (marks 3)**

**Ans.**

1. Complete
2. Correct
3. Feasible
4. Necessary
5. Prioritized
6. Unambiguous
7. Verifiable

**Q: What is Ambiguity? (marks 2)**

ANS: Ambiguity means that two different readers of the same document interpret the requirement differently.

**Q: What is Gold-Plating? (Marks 3)**

Ans: These called cool features. Which are not added in requirements but developers add this. Gold-plating refers to features are not present in the original requirement document and in fact are not important for the end-user but the developer adds

them anyway

thinking that they would add value to the product.

**Q: Write the names of level of requirements: (Marks 3)**

ANS:

1. Business Requirements
2. User Requirements
3. Functional Requirements
4. Non-Functional Requirements

**Q: What is Business Scope? (Marks 2)**

ANS: In which document we write business requirements these called business scope individual

**Q: What is Requirement definition? (Marks 2)**

Ans: In which document we write user requirements these called requirement definition

**Q: What is function specification? (Marks 2)**

Ans: In which document we write functional/nonfunctional requirements these called function specification

**Q: Define Business requirement: (Marks 2)**

User will be able to correct spelling errors in a document efficiently and is will be integrated with he existing system.

**Q: Define User requirement: (Marks 2)**

Finding spelling errors in the document and decide whether to replace each misspelled word with the one chosen from a list of suggested words.

**Q: Define Function requirement: (Marks 3)**

- Find and highlight misspelled words
- Display a dialogue box with suggested replacements.
- Making global replacements.

**Q: Define Non functional requirement: (Marks 2)**

It must be integrated into the existing word-processor which runs on windows platform.

**Lec#05 SUBJECTIVE**

**Q : Write the Use Case Model Components? (marks 3)**

**Ans.**

There are two use case model component

- 1.Cases = case specifies a complete functionality
2. Actors= An actor is an entity that has an interest in interacting with the system. An actor can be a human or some other device or system.

**Q: What is scope? (marks 3)**

ANs: Project scope defines the concept and range of the proposed solution, and limitations

identify certain capabilities that the product will not include. Clarifying the scope and

limitations helps to establish realistic stakeholder's expectations

### **Lec#06 SUBJECTIVE**

**Q :Write the name of Delete Information use case? (marks 2)**

Ans. There are two existing use cases

1.Record Transaction

2.Cancel Transaction

**Q : Write the name of Customer classes? (marks 2)**

Ans. There are two classes.

1. Individual Customer

2. Corporate Customer

**Q: What is Exception? (MARKS 2)**

ANS: The system will not allow a user to delete information that is being used in the system.

The system will not allow a user to delete another user that has subordinates.

**Q: What is Assumption Act? (Marks 3)**

ANS: Deleting information covers a permanent deletion of an entire set of data such as a

commission plan, user, group etc. Deleting a portion of an entire set constitutes modifying the set of data.

Deleted information is not retained in the system.

A user can only delete information that has not been used in the system.

**Q: Write the names of limitations of use case? (Marks 2)**

ANS:

- Usability
- Reliability
- Performance
- Portability
- Access

**What is elaborated Use case? Explain it. (marks 3)**

**Answer:**

After the derivation of the use case model, each use is elaborated by adding detail of interaction between the user and the software system.

An elaborated use case has the following components: Use Case Name, actors, summary, precondition, post-condition, extend, uses, normal course of events, alternative path, exception, assumption.

**Q: UML stands for.....?**

ANS: Unified Modeling Language

### **Lec#07 SUBJECTIVE**

**Q: What is Source? (Marks 3)**

**Ans:** Sources of requirements are the origins from where the corresponding business process is initiated. By this concept, one has to trace from a requirement back to its origins to see who is involved in its initiation.

**Q: What is Sink? (Marks 3)**

ANS: Sink is the consumer of certain information. It is that entity which provides a logical end to a business process. Thus, 'sinks of requirements' is a concept that helps in identifying persons, organizations or external systems that gets certain functionality from the system.

**Q. Write the name Techniques of Logical System Modals? (marks 3)**

**Ans.**

1. User business processes
2. User activities for conducting the business processes
3. Processes that need to be automated
4. Processes which are not to be automated

**Q : What is Business process model? (marks 2)**

**Ans.** The first model that we will look at is called the process model. This model provides a high-level pictorial view of the business process. This model can be used as a starting point in giving the basic orientation to the reader of the document.

**Q: There are three models used write their names: (marks 2)**

**ANs:**

- Business Model
- State transition Model
- Data flow Model

## **Lec#08 SUBEJCTIVE**

**Q: What is STD? (Marks 2)**

STD stands for State transition diagrams. This is another technique to document domain knowledge. This is an easy technique to design a work flow application.

**Q : What Types of International Function Point User's Group (IFPUG)? (marks 2)**

Ans. There are three types

1. External Inputs
2. External Outputs
3. External Inquiry

**Q : What is a data flow model? And explain it. (marks 5)**

Ans.

- ☐ Captures the flow of data in a system.
  - ☐ It helps in developing an understanding of system's functionality.
  - ☐ What are the different sources of data, what different transformations take place on data and what are final outputs generated by these transformations.
  - ☐ It describes data origination, transformations and consumption in a system.
  - ☐ Information is organized and disseminated at different levels of abstraction.
- Thus  
this technique becomes a conduit for top down system analysis and requirements modeling.

**Q : Write the types of shapes of notation. And explain it. (marks 5)**

Ans. There are four shapes of notations

**1. Process**

- ☐ What are different processes or work to be done in the system.
- ☐ Transforms of data.

**2. External Agent**

External systems which are outside the boundary of this system. These are represented using the squares

**3. Data Store**

- ☐ Where data is being stored for later retrieval.
- ☐ Provides input to the process
- ☐ Outputs of the processes may be going into these data stores.

#### 4. Data Flow

- ☐ Where the data is flowing.
- ☐ Represents the movement of the data in a data flow diagram.

#### Q: What is External Inputs? (Marks 2)

Ans: An external input (EI) is an elementary process that processes data or control information that comes from outside the application boundary.

#### Q: What is External Query? (Marks 3)

ANS: An external output (EO) is an elementary process that sends data or control information outside the application boundary. The primary intent of an external output is to present information to a user through processing logic other than, or in addition to, the retrieval of data or control information.

#### Q : What is the difference between DFD and Flow Chart. (marks 5)

Ans.

##### DFD

- Processes on a data flow can operate in parallel.
- Looping and branching are typically not shown.
- Each process path may have a very Different timing.

##### Flow Chart

- Processes on flowcharts are sequential.

- Show the sequence of steps as an algorithm and hence looping and branching are part of flowcharts.

### **Lec#09 SUBJECTIVE**

**Q : Write the name of CRUD Operations? And explain it. (marks 3)**

**Ans.**

These are four operations

- ☐ **Create:** creates data and stores it.
- ☐ **Read:** retrieves the stored data for viewing.
- ☐ **Update:** makes changes in an stored data.
- ☐ **Delete:** deletes an already stored data permanently.

**Q : What is Common Mistakes in Data Flow Diagrams? (marks 2)**

**Ans.**

- There is no input for the process Freeze Member Account
- In a similar manner, the process Create a New Member Account does not produce any output.

### **Lec#10 SUBJECTIVE**

**Q : What is GUI or Graphical user interface? (marks 2)**

**Ans.** Graphical user interface is a computer interface that allows user to interact with a device through graphical elements such as pictures and animations, as opposed to text-based commands.

**Q : What is Motivation for GUI? (marks 3)**

**Ans.**

- ☐ System users often judge a system by its interface rather than its functionality

- ☐ A poorly designed interface can cause a user to make catastrophic errors
- ☐ Poor user interface design is the reason why so many software systems are never Used

**Q : Write the three types of Pitfalls of using GUIs in Functional Specifications. (marks 3)**

**Ans.**

- ☐ UIs distract from business process understanding (what) to interfacing details (how)
- ☐ Unstable requirements cause frequent modifications in UIs
- ☐ An extra work to be done at the requirement level each time a GUI change has to be incorporated

**Q :. What is Prototype? (marks 2)**

**Ans.** Prototyping is yet another technique that can be used to reduce customer dissatisfaction at the requirement stage. A prototype is not the real product. It is rather just a real looking mock-up of what would be eventually delivered and might not do anything useful.

Cs 504 lec no 11

**Q1: what parameters are used to measure and analyze design quality? 5 marks**

Answer:- (Page 71) A software design can be looked at from different angles and different parameters can be used to measure and analyze its quality. These parameters include efficiency, compactness, reusability, and maintainability. A good design from one angle may not seem to be suitable when looked from a different perspective. For example, a design that yields efficient and compact code may not be very easy to maintain. In order to establish whether a particular design is good or not, we therefore have to look at the project and application requirements

Question No: 2 What should be consideration for maintain design? (Marks: 5)

Answer:- (Page 71) In order to make a design that is maintainable, it should be understandable and the changes should be local in effect. That is, it should be such that a change in some part of the system should not affect other parts of the system. This is achieved by applying the principles of modularity, abstraction, and separation of concern. If applied properly, these principles yield a design that is said to be more cohesive and loosely coupled and thus is easy to maintain.

Question No: 3 It is fact that good design makes maintenance easier. Which design principle help this to be achieved? ( Marks: 3 )

Answer:- (Page 71) A good design from one angle may not seem to be suitable when looked from a different perspective. For example, a design that yields efficient and compact code may not be very easy to maintain. In order to establish whether a particular design is good or not, we therefore have to look at the project and application requirements.

Q 4 :To manage the complexity of the system we need to apply the principles of separation of concern. Discuss briefly 2 MARKS

Answer: (Page 69) Separation of concern allows us to deal with different individual aspects of a problem by considering these aspects in isolation and independent of each other. A complex system may be divided into smaller pieces of lesser complexity called modules.

Q5: DEFINE Software Design Qualities ?

Ans : A software design can be looked at from different angles and different parameters can be used to measure and analyze its quality. These parameters include efficiency, compactness, reusability, and maintainability.

Q6 : what is data modeling ? 2 marks

Ans : s. Data modeling is an essential activity performed during the design phase. This includes the identification of data entities and their attributes, relationships among these entities, and the appropriate data structures for managing this data.

## Lecture no 12

Q1:Define abstraction? 2 marks

Answer:- (Page 79) An abstraction is a technique in which we construct a model of an entity based upon its essential characteristics and ignore the inessential details.

Question No: 2 ( Marks: 3 ) To manage the complexity of the system we need to apply the principle of abstraction. Discuss briefly?

Answer:- (Page 79) An abstraction is a technique in which we construct a model of an entity based upon its essential characteristics and ignore the inessential details. The principle of abstraction also helps us in handling the inherent complexity of a system by allowing us to look at its important external characteristic, at the same time, hiding its inner complexity. Hiding the internal details is called encapsulation.

Q 3 :What are architectural designs Process, explain briefly? 5 Marks

Answer:- (Page 79) System structuring: - System structuring is concerned with decomposing the system into interacting sub-systems. The system is decomposed into several principal sub-systems and communications between these sub-systems are identified. Control modeling:- Control modeling establishes a model of the control relationships between the different parts of the system. Modular decomposition:- During this activity, the identified sub-systems are decomposed into modules. This design process is further elaborated in the following section where architectural views are discussed.

Q4:Define cohesion 2 marks

. Answer: (Page 72) Cohesion is an internal property of a module. Cohesion describes the intra-component linkages while couple shows the inter-component linkages. Cohesion measures the independence of a module

Q 5:Define coupling ? 2 marks

Ans : Coupling is a measure of independence of a module or component. Loose coupling means that different system components have loose or less reliance upon each other.

Q 6:What is encapsulation? 2 marks

Ans : Hiding the internal details is called encapsulation

Question No: 7 ( Marks: 5 ) What is action-oriented approach for Software Design?

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Answer: (Page 80) In the case of action-oriented approach, data is decomposed according to functionality requirements. That is, decomposition revolves around function. In the OO approach, decomposition of a problem revolves around data. Action-oriented paradigm focuses only on the functionality of a system and typically ignores the data until it is required. Object-oriented paradigm focuses both on the functionality and the data at the same time. The basic difference between these two is decentralized control mechanism versus centralized control mechanism respectively. Decentralization gives OO the ability to handle essential complexity better than action-oriented approach.

### Lecture no 13

Question No: 1 ( Marks: 3 ) HOW DO YOU DETERMINE THAT AN OBJECTIVE BELONGS TO CERTAIN CLASS?

Answer:- (Page 85) The basic unit of object oriented design is an object. An object can be defined as a tangible entity that exhibits some well defined behavior. The structure and behavior of similar objects are defined in their common class. A class specifies an interface and defines an implementation.

Q2: What is behavior driven perceptive of an objective? 3 marks

Answer:- (Page 85) Behavior is how an object acts and reacts in terms of its state changes and message passing. The behavior of an object is completely defined by its actions. A message is some action that one object performs upon another in order to elicit a reaction. The operations that clients may perform upon an object are called methods

Q3: What is the difference between Aggregation and Association? 3 marks

Answer:- (Page 87) As compared to association, aggregation implies a tighter coupling between the two objects which are involved in this relationship. Therefore, one way to differentiate between aggregation and association is that if the two objects are tightly coupled, that is, if they cannot exist independently, it is an aggregation, and if they are usually considered as independent, it is an association.

Q4 : define Relationship Among Objects ? 2 marks

The object model presents a static view of the system and illustrates how different objects collaborate with one another through patterns of interaction. Inheritance, association and aggregation are the three inter-object relationships specified by the object model.

Q5 :Define the object model?

The elements of object oriented design collectively are called the Object Model. The object model encompasses the principles of abstraction, encapsulation, and hierarchy or inheritance.

Question No: 25 ( Marks: 5 ) Code example of High Coupling

Answer: Click here for detail Tightly Coupled Example: 

```
public class CartEntry { public float Price; public int Quantity; } public class CartContents { public CartEntry[] items; } public class Order { private CartContents cart; private float salesTax; public Order(CartContents cart, float salesTax) { this.cart = cart; this.salesTax = salesTax; } public float OrderTotal() { float cartTotal = 0; for (int i = 0; i < cart.items.Length; i++) { cartTotal += cart.items[i].Price * cart.items[i].Quantity; } cartTotal += cartTotal*salesTax; return cartTotal; } }
```

## Lecture no 14

### Q1: What is Textual Analysis? Explain it 3 marks

Answer:- (Page 90) Textual analysis was developed by Abbot and then extended by Graham and others. In this technique different parts of speech are identified within the text of the specification and these parts are modeled using different components.

### Q2 : name the four layers of the OO design pyramid ? 2 marks

The four layers of the OO design pyramid are:

- 1) The subsystem layer.
- 2) The class and object layer
- 3) The message layer
- 4) The responsibility layer

### Q 3: define problem statement? 3 marks

A simple cash register has a display, an electronic wire with a plug, and a numeric keypad, which has keys for subtotal, tax, and total. This cash storage device has a total key, which triggers the release on the drawer..

### Q4: Explain the four layer of the OO design pyramid ? 5 marks

1) The subsystem layer. Contains a representation of each of the subsystems that enable the software to achieve its customers defined requirements and to implement the technical infrastructure that supports customer requirements.

2) The class and object layer. Contains the class hierarchies that enable the system to be created using generalization and increasingly more targeted specializations. The layer also contains design representations for each object.

3) The message layer. Contains the details that enable each object to communicate with its collaborators. This layer establishes the external and internal interfaces for the system

. 4) The responsibility layer. Contains the data structures and algorithmic design for all attributes and operations for each object.

## Lecture no 16

### No: 1 ( Marks: 5 ) How the objects are identified in peter codd's technique?

**Answer: (Page 93)** Objects are identifying in the following way.

**Select actors** : Actors are people and organizations that take part in the system under consideration. Examples of actors are: person, organization (agency, company, corporation, foundation)

**. Select Participants** A participant is a role that each actor plays in the system under consideration. Examples of participants are: agent, applicant, buyer, cashier, clerk, customer, dealer, and distributor. Etc.

**Select Places** :Places are where things come to rest or places that contain other objects. Examples of places are: airport, assembly-line, bank, city, clinic, country, depot, garage and hospital etc

**. Select Transactions** : Transactions are the "events". These transactions usually come from a window (GUI), some object which monitors for significant event and logs that information, or a another system that interacts with the system under consideration and logs some information. Examples of transactions are: agreement, assignment, authorization, contract, delivery, deposit, incident, inquiry, order, payment, problem report, purchase and sales etc.

**Select Container Objects** Containers are objects that hold other objects. e.g. bin, box, cabinet, folder, locker, safe, shelf, etc. Therefore a place is also a container but every container need not be a place.

**Select Tangible things** Take a "walk" through the system and select "tangible" things around you used in the problem domain. These may be characterized as all the remaining (not yet selected) "nouns" that make up the problem domain. Examples are: account, book, calendar, cash box, cash drawer, item, plan, procedure, product, schedule, skill, tool, etc

## Lecture no 17

### Q1: what is Identify Structures?

**Ans** : A structure is a manner of organization which expresses a semantically strong organization within the problem domain.

### Q2: how many types of Identify Structures?

**Ans** : There are two type of structures

Generalization-Specialization (Gen-Spec) and whole-part.

### Q3 : Define Attributes - ?

**Ans** : The first two activities would identify most of the objects (classes) in the problem domain. Now is the time to think about the role and responsibilities of these objects. The first thing to consider is their attributes,

### Q4 : Show Collaborations (associations and aggregations) - Who I

## know? 5 marks

The second step in establishing each object's responsibility is to identify and show how this object collaborates with other objects, i.e., who it knows. These collaborations can be identified with the help of the following steps:

1. For an actor, include an object connect to its participants (association).
2. For a participant, include an object connection to its actor (already established) and its transactions (association).
3. For a location, include object connections to objects that it can hold (association), to its part objects (aggregation), and to the transactions that are taking place at that location (association).
4. For transactions, include object connections to its participants (already established), its line items (aggregation), and its immediate subsequent transaction (aggregation).
5. For a transaction line item, include object connections to its transaction (already established), its item (association), a companion

Q5 : GIVE EXAMPLE ARE ATTRIBUTES ?

Ans : Examples of attributes are: number, name, address, date, time, operational state, phone, status, threshold, type, etc

## lecture no 19

**Q 1: Keeping in mind the Connie's case study, what rule of thumbs was identified, list them down ? 5 marks**

**Ans :**

Who I Know - Rules of

Thumb

1 :

an actor knows about its participants person knows about cashier

2 :

a transaction knows about its participants a session knows about its register and cashier

3

A transaction contains its transaction line items sale contains its sales line items

4

A transaction knows its sub transactions session knows about its sales sale knows about its payments

5

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A place knows about its transactions store knows about its sessions

6

A place knows about its descriptive objects store knows about its tax categories

7

A container knows about its contents a store knows about its cashiers, items, and registers

Q2:

**Keeping connie's case study in mind, as discussed in lecture, list down whole parts which were identified. Answer: (Page 100) 3 marks**

Identify Whole-Part Structures

1 :

1 :A store as a whole is made up of cashiers, registers, and items.

2 :PA register contains a cash drawer.

3: A sale is constituted of sale line items.

Q3 :

Define data flow diagram ? 2 marks

**Answer: (Page 100)** A data flow diagram (DFD) is a graphical representation of the data through an information system, modeling its process aspects. Often they are a preliminary step used to create an overview of the system which can later be elaborated.

## Lecture no 20 :

### Q1: Why we use series of diagram? 3 marks

A series of diagrams can be used to describe the *dynamic behavior* of an object-oriented

system. This is done in terms of a set of messages exchanged among a set of objects within a context to accomplish a purpose.

**Q2 : Define the purpose of Interaction diagrams is to ? 3 marks**

The purpose of Interaction diagrams is to:

- Model interactions between objects
- Assist in understanding how a system (a use case) actually works
- Verify that a use case description can be supported by the existing classes
- Identify responsibilities/operations and assign them to classes

**Q3 : Define life line of objects ? 2 marks**

Ans The boxes denote objects (or classes), the solid lines depict messages being sent from one object to the other in the direction of the arrow, and the dotted lines are called life-lines of objects..

**q4 : write the syntax used for naming objects in a sequence diagram ? 3 marks**

The syntax used for naming objects in a sequence diagram is as follows:

- syntax: [instanceName][:className]
- Name classes consistently with your class diagram (same classes).
- Include instance names when objects are referred to in messages or when several

## **Lecture No 21:**

**(Q) How Many Types of messages are there?**

Ans. There Are four type of messages are there.

(1) Synchronous

(2) Asynchronous

(3) Create

(4) Destroy

**Q No: 2. what is *Synchronous Messages*?**

Synchronous messaging describes communications that takes place between two applications or systems, where the system places a message in a message queue (also called an Event Queue in enterprise messaging systems) and then waits for a message response before it continues processing. *Contrast with asynchronous messaging. They are denoted by the full arrow.*

Synchronous messaging is also known as synchronous communication.

**(Q3) What is *Asynchronous Messages*?**

Ans.

Asynchronous messages are "signals," denoted by a half arrow. They do not block the caller. Asynchronous messages typically perform the following actions:

Create a new thread.

Create a new object.

Communication with thread is already running.

**Q4. *Object Creation and Destruction***

Ans. An object may create another object via a <<create>> message. Similarly an object may Destroy another object via a <<destroy>> message. An object may also destroy itself. One Should avoid modeling object destruction unless memory management is critical.

**Q: No 5. Collaboration Diagrams depict Dynamic behavior of the system, explain it.**

Ans.

Collaboration diagrams can also be used to depict the dynamic behavior of a system. They show how objects interact with respect to organizational units (boundaries!). Since a boundary shapes communication between system and outside world e.g. user interface or other system, collaboration diagrams can be used to show this aspect of the system. The sequence of messages determined by numbering such as 1, 2, 3, 4, This shows which operation calls which other operation.

Collaboration diagrams have basically two types of components: objects and messages.

Objects exchange messages among each-other. Collaboration diagrams can also show synchronous, asynchronous, create, and destroy message using the same notation as used

In sequence diagrams. Messages are numbered and can have loops

**Q: No 6. Evaluating the Quality of an Object-Oriented Design.**

Ans. Judging the quality of a design is difficult. We can however look at certain object-Oriented design attributes to estimate its quality. The idea is to analyze the basic principle

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Of encapsulation and delegation to judge whether the control is centralized or distributed, Hence judging the coupling and cohesion in a design. This will tell us how maintainable a Design is.

You may also recall our earlier discussion of coupling and cohesion. It can be easy to see

That OO design yield more cohesive and loosely coupled systems.

## Lecture No 22:

### Q: No 1. What is Software Architecture?

Ans.

**Software architecture** refers to the fundamental structures of a [software system](#), the discipline of creating such structures, and the documentation of these structures. These structures are needed to reason about the software system. Each structure comprises software elements, relations among them, and properties of both elements and relations,<sup>[1]</sup> along with [rationale](#) for the introduction and configuration of each element. The *architecture* of a software system is a metaphor, analogous to the [architecture](#) of a building.

Q: No 2.

### Ans. Why is architecture important?

If a project has not achieved a system architecture, including its rationale, the Project should not proceed to full-scale system development. Specifying the Architecture as a deliverable enables its use throughout the development and Maintenance process.

**Q: 3. No Architectural design process**

Ans. Just like any other design activity, design of software architecture is a creative and Iterative process. This involves performing a number of activities, not necessarily in any Particular order or sequence. These include system structuring, control modeling, and Modular decomposition.

**Q: No 4. Architectural Attributes**

Ans. Software architecture must address the non-functional as well as the functional Requirements of the software system. This includes performance, security, safety, Availability, and maintainability.

**Q: No 5. What Is Performance in Architectural Design?**

Ans. – Performance can be enhanced by localizing operations to minimize sub-System communication. That is, try to have self-contained modules as Much as possible so that inter-module communication is minimized.

**Q: No 6. What is Security in Architectural Design?**

Ans. – Security can be improved by using a layered architecture with critical Assets put in inner layers.

**Q: No. 7. what is Safety in Architectural design**

Ans. – Safety-critical components should be isolated

**Q: No what is Availability in Architectural design**

Ans. – Availability can be ensured by building redundancy in the system and Having redundant components in the architecture.

**Q: No 8. What is Maintainability in Architectural design?**

Ans. – Maintainability is directly related with simplicity. Therefore, Maintainability can be increased by using fine-grain, self-contained Components.

**Q: No 9. What are architectural designs Process, explain briefly?**

Ans. Just like any other design activity, design of software architecture is a creative and Iterative process. This involves performing a number of activities, not necessarily in any

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Particular order or sequence. These include system structuring, control modeling, and

Modular decomposition.

**System structuring:** - System structuring is concerned with decomposing the system into interacting sub-systems. The system is decomposed into several principal sub-systems and communications between these sub-systems are identified.

**Control modeling:** - Control modeling establishes a model of the control relationships between the different parts of the system.

**Modular decomposition:** - During this activity, the identified sub-systems are decomposed into modules. This design process is further elaborated in the following section where architectural views are discussed.

**Q: No 10. Differentiate between architectural design and system architecture in a single line?**

Ans. Architecture faces towards strategy, structure and purpose, towards the abstract; implementation and practice, towards the concrete.

Team Work

Question No : 1 of 26

Marks: 1 (Budgeted Time 1 Min)

A change becomes ----- because of close presence of data and functions

Answer ( Please select your correct option )

☒ Localized

Correct answer solved by hadi  
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☐ Private

☐ Global

☐ Accessible

**Made By: Waqar Siddhu**

Question No : 2 of 26

Marks: 1 (Budgeted Time 1 Min)

'A website developed to sell movies over the internet', is \_\_\_\_\_.

Answer ( Please select your correct option )

☐

Scientific software

☐

Safety critical software

☒

Web based software

☐

Embedded software

Correct answer solved by hadi  
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Email: usmanraj20@gmail.com

**Made By: Waqar Siddhu**

Question No : 3 of 26

Marks: 1 (Budgeted Time 1 Min)

Given below are some statements associated with prototyping. Which one is correct statement?

Answer ( Please select your correct option )

☐ The scope and the complexity of the system can quickly expand beyond original plans.

☐ Prototyping decreases creativity since it does not allow quick feedback.

☐ Prototypes negate the requirement for the system analysis phase.

☒ Prototyping is a technique that can be used to reduce customer satisfaction

**Made By: Waqar Siddhu**

Question No : 4 of 26

Marks: 1 (Budgeted Time 1 Min)

Which of the following is not a fact finding method?

Answer ( Please select your correct option )

☐

Site visits

☒

Prototyping

Correct answer solved by hadi  
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☐

Study of similar systems

☐

Modulation

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Question No : 5 of 26

Marks: 1 (Budgeted Time 1 Min)

The nature of software applications can be characterized by their information \_\_\_\_\_

Answer ( Please select your correct option )

☒

Complexity

☐

Content

☐

Determinacy

☐

Content and determinacy

**Made By: Waqar Siddhu**

Question No : 6 of 26

Marks: 1 (Budgeted Time 1 Min)

A project is considered successful if:

Answer ( Please select your correct option )

- ☒ The system was delivered in time and with in budget
- ☐ The system meets at least some of the customer's requirements.
- ☐ The system development process has a maximum impact on the ongoing business process..
- ☐ Minimum time was spent for requirement gathering and designing.

Correct answer solved by hadi  
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**Made By: Waqar Siddhu**

Question No : 7 of 26

Marks: 1 (Budgeted Time 1 Min)

Cohesion is an internal property of a module whereas coupling is its----- with other modules

Answer ( Please select your correct option )

☐ Relationships

☒ Dependence

☐ Independence

☐ Separation

**Made By: Waqar Siddhu**

Question No : 8 of 26

Marks: 1 (Budgeted Time 1 Min)

As per Peter Coad's methodology the following may NOT be a perfect candidate for being an object

Answer ( Please select your correct option )

☐

Folder

☐

Bin

☐

Student

☐

Size

Maybe

Made By: Waqar Siddhu

Question No : 9 of 26

Marks: 1 (Budgeted Time 1 Min)

\_\_\_\_\_ requirements are often called product features.

Answer ( Please select your correct option )



Functional

Correct answer solved by hadi  
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Business



User



Non-functional

**Made By: Waqar Siddhu**

Question No : 10 of 26

Marks: 1 (Budgeted Time 1 Min)

Function oriented methodology is also known as \_\_\_\_\_.

Answer ( Please select your correct option )

☐ Object Oriented

☐ Structure Oriented

☒ Action Oriented

☐ Module Oriented

**Made By: Waqar Siddhu**

Question No : 11 of 26

Marks: 1 (Budgeted Time 1 Min)

Software crisis appeared in \_\_\_\_\_.

Answer ( Please select your correct option )

☒ Early 50s

Correct answer solved by hadi  
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Email: usmanraj20@gmail.com

☐ Early 60s

☐ Early 70s

☐ Early 80s

**Made By: Waqar Siddhu**

Question No : 12 of 26

Marks: 1 (Budgeted Time 1 Min)

\_\_\_\_\_ is another technique to document domain knowledge.

Answer ( Please select your correct option )

☐

Business model

☐

Activity diagram

☐

Use-case model

☒

State transition diagram

**Made By: Waqar Siddhu**

Question No : 13 of 26

Marks: 1 (Budgeted Time 1 Min)

Agreement, assignment, authorization, contract, delivery, deposit, incident etc are examples of \_\_\_\_\_.

Answer ( Please select your correct option )

☐

Object

☐

Participant

☐

Tangible

☒

Transaction

**Made By: Waqar Siddhu**

Question No : 14 of 26

Marks: 1 (Budgeted Time 1 Min)

Oval shape in state transition diagram represents a \_\_\_\_\_.

Answer ( Please select your correct option )



State

Correct answer solved by hadi  
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Email: usmanraj20@gmail.com



Input



Output



Data

**Made By: Waqar Siddhu**

Question No : 15 of 26

Marks: 1 (Budgeted Time 1 Min)

In a DFD, communication between two external agents without an intermediate process is \_\_\_\_\_.

Answer ( Please select your correct option )

☒ Prohibited

☐ Must

☐ Possible

☐ Avoided

**Made By: Waqar Siddhu**

Question No : 16 of 26

Marks: 1 (Budgeted Time 1 Min)

UML provides \_\_\_\_\_ different mechanisms to document the dynamic behavior of the system.

Answer ( Please select your correct option )



Two

Correct answer solved by hadi  
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Email: usmanraj20@gmail.com



Three



Four



Five

**Made By: Waqar Siddhu**

Question No : 17 of 26

Marks: 1 (Budgeted Time 1 Min)

Diagrams used to capture dynamic behavior are called \_\_\_\_\_ diagrams.

Answer ( Please select your correct option )



Interaction



Flow



Sequence



Collaboration

**Made By: Waqar Siddhu**

Question No : 18 of 26

Marks: 1 (Budgeted Time 1 Min)

In sequence diagram, a message is represented by a/an \_\_\_\_\_.

Answer ( Please select your correct option )



Arrow

Correct answer solved by hadi  
Cell No:03228043306  
Email: usmanraj20@gmail.com

☐

Straight line

☐

Dotted line

☐

Box

**Made By: Waqar Siddhu**

Question No : 19 of 26

Marks: 1 (Budgeted Time 1 Min)

\_\_\_\_\_ is NOT a type of messages of sequence diagram

Answer ( Please select your correct option )

☐ Synchronous

☐ Asynchronous

☐ Create

☒ Update

ver 2/19/11

**Made By: Waqar Siddhu**

Question No : 20 of 26

Marks: 1 (Budgeted Time 1 Min)

UML(Unified Modeling Language) is among different ..... which are used for documenting the object oriented design.

Answer ( Please select your correct option )

☐ Documents

☒ Notations

☐ SRS

☐ Reports

Correct answer solved by hadi  
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Email: usmanraj20@gmail.com

**Made By: Waqar Siddhu**

Question No : 21 of 26

Marks: 2 (Budgeted Time 4 Min)

You are asked to develop object oriented design for a system. Your concern about decomposition would be based on function or data ?

Answer ( Please [click here](#) to Add Answer )

Normal Arial 12 B I U

*both*

**Made By: Waqar Siddhu**

Question No : 22 of 26

Marks: 2 (Budgeted Time 4 Min)

Early design decisions can vary in software development. Do you agree or not? Justify your choice.

Answer ( [Please click here to Add Answer](#) )

Normal Arial 12 B I U

**Made By: Waqar Siddhu**

Question No : 23 of 26

Marks: 3 (Budgeted Time 6 Min)

Ali is considered as an object. What are the characteristics which decide Ali is an Object?

Answer ( [Please click here to Add Answer](#) )

Normal Arial 12 B I U

**Made By: Waqar Siddhu**

Question No : 24 of 26

Marks: 3 (Budgeted Time 6 Min)

Differentiate between a sub-system and module.

Answer ( [Please click here to Add Answer](#) )

Normal Arial 12 B I U

**Made By: Waqar Siddhu**

Question No : 25 of 26

Marks: 5 (Budgeted Time 10 Min)

In textual analysis we analyze the text to identify the classes, objects and relationships among objects and classes. What are the meanings of these terms in textual analysis?

- 0. Proper Noun
- 1. Improper Noun
- 2. Being Verb
- 3. Having Verb

Answer ( [Please click here to Add Answer](#) )



**Made By: Waqar Siddhu**

Question No : 26 of 26

Marks: 5 (Budgeted Time 10 Min)

Using the architecture of a house or building, draw a comparison with software architecture.

Answer ( [Please click here to Add Answer](#) )



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